AerE 421 - Lab Assignment 2

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Introduction – Part 1

For this week's lab, we were assigned two separate exercises. First, we modeled into Ansys the main spar of an aircraft subjected to a lift and drag force to collect the y and z displacements, bending stresses, point stresses, and axial stresses. It is assumed that the spar takes the entire uniform distributed load of 250 N/m. The spar is modeled as an I-beam fixed at one end and free at the other with dimensions length L = 2.5m, depth d = 0.09m, width b = 0.05m, web thickness t_w = 0.001m, and flange thickness t_f = 0.002 m. The beam is made of an isotropic linear elastic material with Young's modulus E = 70 GPa and Poisson's ratio v = 0.3.

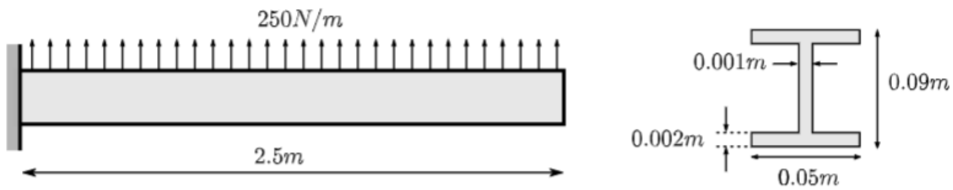

Figure 1. *Dimensions and load distribution of modeled main spar of an aircraft.*

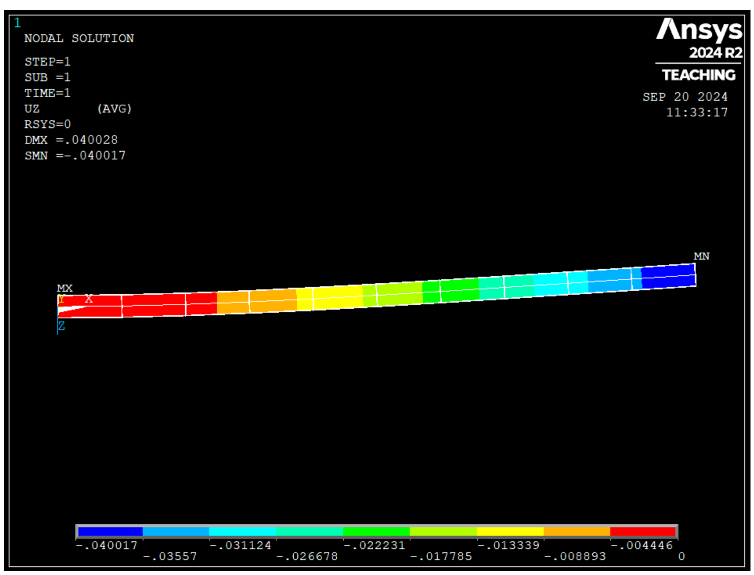

Figure 2. *Top view of contour plot of the beam; Z displacement of beam (m)*

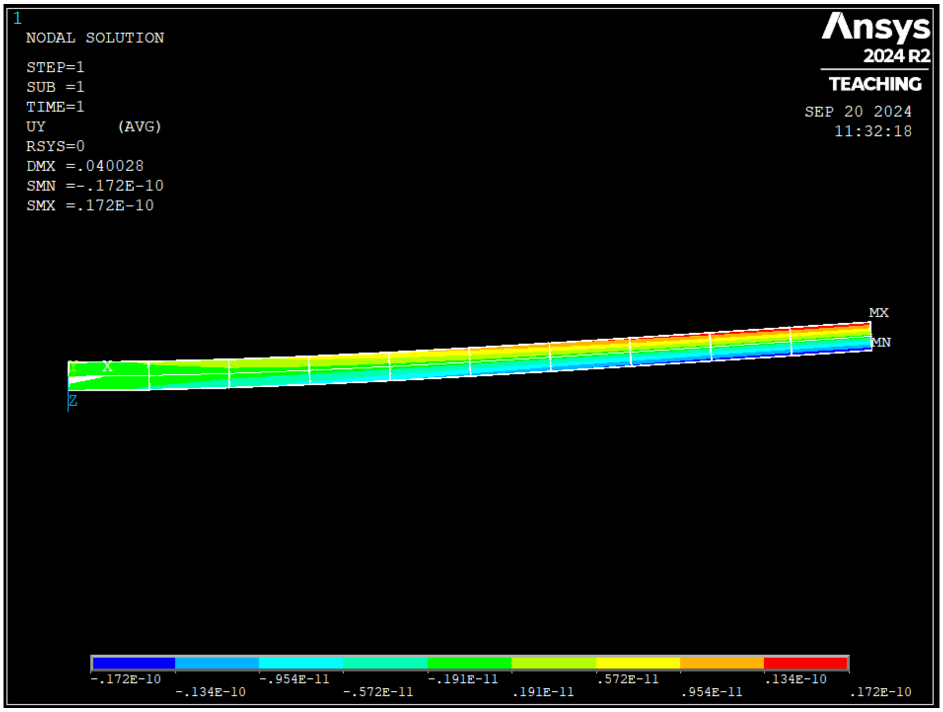

Figure 3. *Front view of contour plot of the beam; Y displacement of the beam (m)*

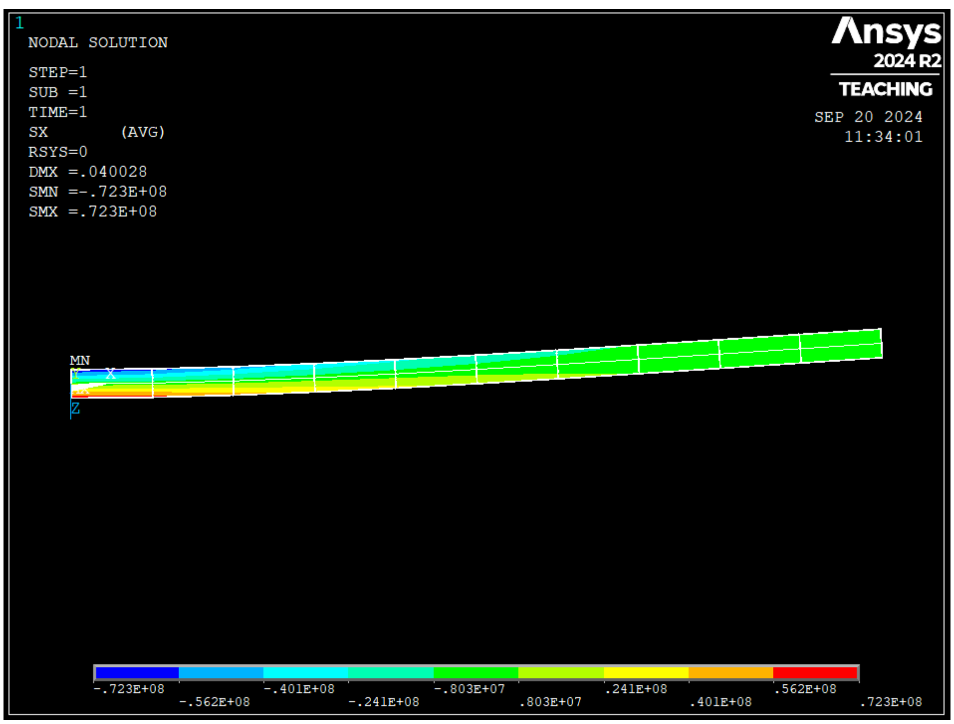

Figure 4. *Top view of contour plot of the beam; X component of stress (N/m^2)*

Table 1. *Table of element stresses, axial stresses, and bending stresses*

Element n⁰	Element Stress (N/m^2)	Axial Stress (N/m^2)	Bending Stress (N/m^2)
1	-72265448	0.0000	0.19931e-8
2	-57892212	0.0000	0.15967e-8
3	-45116000	0.0000	0.12444e-8
4	-33936812	0.0000	0.93605e-9
5	-24354654	0.0000	0.67177e-9
6	-16369522	0.0000	0.45152e-9
7	-9981416	0.0000	0.27533e-9
8	-5190336	0.0000	0.14317e-9
9	-0.19963e7	0.0000	0.55071e-10
10	-0.39926e6	0.0000	0.11019e-10

Discussion

To solve for the Euler-Bernoulli beam theory, the moment of inertia was calculated I = 4.403e-7 m^3. With an online beam calculator that uses the Euler-Bernoulli beam theory as the solver, we could collect displacement values to compare to those output by Ansys, resulting in the following graph:

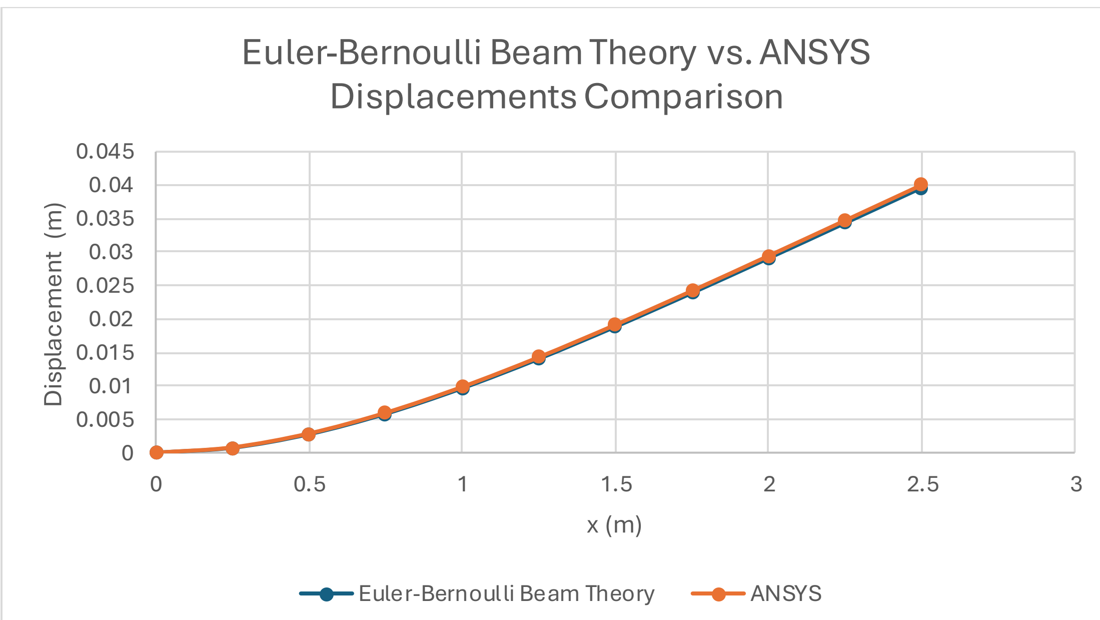

Figure 5. *Euler-Bernoulli Beam Theory vs. ANSYS Displacements Comparison Graph*

It becomes evident from observing the graph that the ANSYS solution and the results of the Euler-Bernoulli beam theory output the same result.

Introduction – Part 2

For part 2, we were prompted to model a landing gear with applied load at the wheels in ANSYS. We modeled a vertical main member OAB and two two-force members AC and AD. As seen in the schematic below, point B acted as a fixed end, points C and D acted as pinned joints, and point A connected all three beams as a pinned joint.

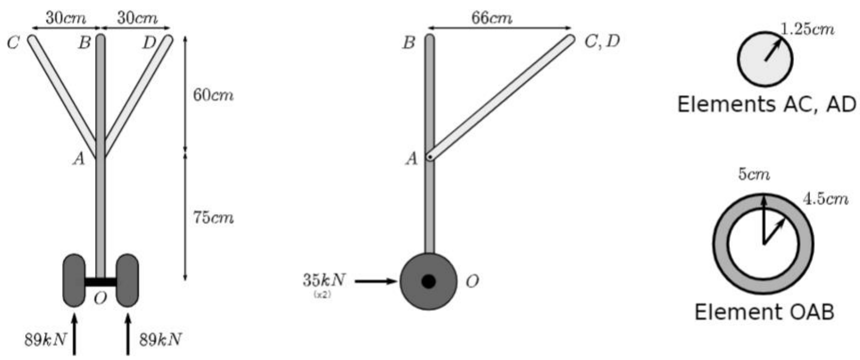

Figure 6. *Dimensions and applied loads for modeled landing gear*

Collected Data

PRINT U NODAL SOLUTION PER NODE

***** POST1 NODAL DEGREE OF FREEDOM LISTING *****

LOAD STEP= 1 SUBSTEP= 1
TIME= 1.0000 LOAD CASE= 0

THE FOLLOWING DEGREE OF FREEDOM RESULTS ARE IN THE GLOBAL COORDINATE SYSTEM

NODE	UX	UY	UZ	USUM
1	0.0000	0.44761E-003	-0.53164E-001	0.53166E-001
2	0.0000	0.40436E-003	-0.46626E-001	0.46627E-001
3	0.0000	0.36111E-003	-0.40171E-001	0.40173E-001
4	0.0000	0.31786E-003	-0.33887E-001	0.33888E-001
5	0.0000	0.27461E-003	-0.27856E-001	0.27857E-001
6	0.0000	0.23135E-003	-0.22164E-001	0.22165E-001
7	0.0000	0.18810E-003	-0.16895E-001	0.16896E-001
8	0.0000	0.14485E-003	-0.12135E-001	0.12136E-001
9	0.0000	0.10160E-003	-0.79670E-002	0.79677E-002
10	0.0000	0.58347E-004	-0.44768E-002	0.44772E-002
11	0.0000	0.15095E-004	-0.17488E-002	0.17488E-002
12	0.0000	0.13585E-004	-0.33337E-003	0.33365E-003
13	0.0000	0.12076E-004	0.61267E-003	0.61279E-003
14	0.0000	0.10566E-004	0.11620E-002	0.11620E-002
15	0.0000	0.90570E-005	0.13872E-002	0.13872E-002
16	0.0000	0.75475E-005	0.13610E-002	0.13610E-002
17	0.0000	0.60380E-005	0.11559E-002	0.11559E-002
18	0.0000	0.45285E-005	0.84466E-003	0.84467E-003
19	0.0000	0.30190E-005	0.49983E-003	0.49984E-003
20	0.0000	0.15095E-005	0.19407E-003	0.19408E-003
21	0.0000	0.0000	0.0000	0.0000
31	0.0000	0.0000	0.0000	0.0000
41	0.0000	0.0000	0.0000	0.0000

MAXIMUM ABSOLUTE VALUES

NODE	0	1	1	1
VALUE	0.0000	0.44761E-003	-0.53164E-001	0.53166E-001

PRINT ELEMENT TABLE ITEMS PER ELEMENT

***** POST1 ELEMENT TABLE LISTING *****

STAT ELEM	CURRENT MEMFORCE	CURRENT AXSTRESS
43	-178000.	-0.11938E+009
44	-178000.	-0.11938E+009
45	-178000.	-0.11938E+009
46	-178000.	-0.11938E+009
47	-178000.	-0.11938E+009
48	-178000.	-0.11938E+009
49	-178000.	-0.11938E+009
50	-178000.	-0.11938E+009
51	-178000.	-0.11938E+009
52	-178000.	-0.11938E+009
53	-7765.3	-0.52078E+007
54	-7765.3	-0.52078E+007
55	-7765.3	-0.52078E+007
56	-7765.3	-0.52078E+007
57	-7765.3	-0.52078E+007
58	-7765.3	-0.52078E+007
59	-7765.3	-0.52078E+007
60	-7765.3	-0.52078E+007
61	-7765.3	-0.52078E+007
62	-7765.3	-0.52078E+007
63	-0.13350E+006	-271896889.
64	-0.13350E+006	-271896889.

MINIMUM VALUES

ELEM	44	63
VALUE	-0.17800E+006	-0.27190E+009

MAXIMUM VALUES

ELEM	53	57
VALUE	-7765.3	-0.52078E+007

***Figure 7**: The first table details each element displacement, while the second displays the member forces and axial stresses present in all members of the system.*

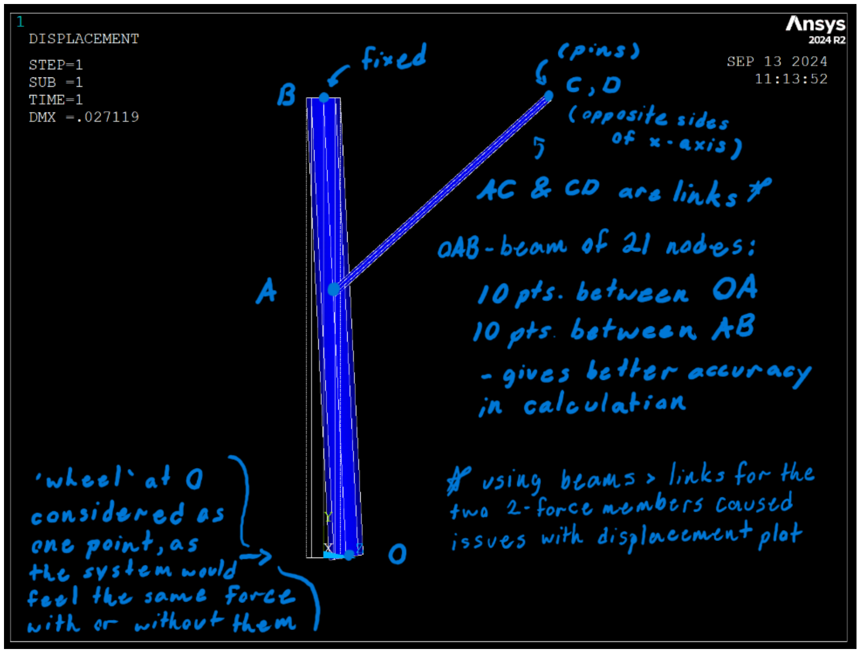

***Figure 8**: Side view of deformed model with design choices (expanded upon below)*

Discussion

For the member OAB, we modeled it as a series of 20 two-node beam elements with a hollow-circular cross section of inner radius r_i = 0.045 m and outer radius r_o = 0.05 m. For members AC and AD, we modeled two links between the nodes A and C, and A and D. We inserted an area A = 4.90874e-4 m^2, derived from the area of a circle with radius r = 0.0125 m.

By means of simplification, the wheels were modeled as a single point O. Thus, the forces acting on them were summed into two resultant vectors in the Y and Z directions, F_Y = 178,000 N and F_Z = -70,000 N. These summed forces were applied as point loads at O.

Taking the 'max' value from the table of element stresses and using the given yield strength of 1482 MPa allows us to calculate the factor of safety of the system using:

$$FOS = \frac{Y_s}{\sigma} = \frac{1482 \cdot 1000000}{|-0.27190 \cdot 10^9|} = 5.45$$

This seems a bit high for our field of study, which may be due to some issues we had during the lab seeing elements disappear after we modelled them, which is why the forces and stresses table has elements in the 60s.

Sources

SkyCiv Engineering. (n.d.). Free Moment of Inertia Calculator. Retrieved from https://skyciv.com/free-moment-of-inertia-calculator/

WebStructural. (n.d.). *Beam Calculator*. Retrieved from https://webstructural.com/beam-calculator.html